

# KALAM CX

## Data Schema & Entity Relationship Design (ERD)

*AI-Powered After-Call Automation Platform*

|                |                                                |
|----------------|------------------------------------------------|
| Project Name   | AI-Powered After-Call Automation Platform      |
| Document Title | Data Schema & Entity Relationship Design (ERD) |
| Version        | 1.0                                            |
| Prepared For   | Kalam CX                                       |
| Date           | July 2026                                      |
| Document Owner | Enterprise Data Architecture Team              |

# 1. Database Overview

## Purpose

The data schema exists to support a reliable, auditable, and non-hallucinating record of every customer support call handled through the AI-Powered After-Call Automation Platform. It provides the structural foundation for customer lookup, documentation retrieval, AI-based extraction, classification, business rule validation, human review, final storage, and operational analytics.

## Scope

The schema covers the full lifecycle of a single call, from the moment a completed transcript enters the system through final approved storage and its inclusion in aggregated analytics. It spans customer and agent identity, call metadata, AI-generated documentation, ticketing, human review, reference/lookup data, and reporting.

## Goals

- Guarantee that every stored fact can be traced back to either the transcript, a human reviewer, or an explicit business rule.
- Prevent fabricated or inferred data by using the explicit value Not Stated wherever information is unavailable.
- Provide a Human-in-the-Loop control point for every generated record prior to final approval.
- Support multi-tenant operation across many client companies such as Kalam CX and its customers.
- Enable operational analytics without degrading transactional performance.

## Why the Schema Is Normalized

The schema is organized in Third Normal Form (3NF) so that each fact is stored exactly once. Reference data such as Categories, Sentiments, and Dispositions is separated from transactional data such as CallRecords, eliminating update anomalies and ensuring classification values remain consistent across the platform.

## Scalability, Maintainability & Auditing

Normalization keeps transactional tables narrow while a dedicated OperationalStatistics table absorbs reporting cost. Surrogate keys, targeted indexing, and tenant partitioning through ClientCompanies allow horizontal scale. Clear entity boundaries mean changes to one area, such as adding a disposition, need no structural changes elsewhere. Every CallRecords row carries created\_at/updated\_at timestamps, and every review action is logged in ReviewQueue with reviewer identity and edited fields — together preserving a complete, reconstructable history.

# 2. Database Design Principles

## Normalization

Entities are decomposed so non-key attributes depend only on their own table's primary key, eliminating redundant storage of customers, agent, and reference data.

## Referential Integrity

Every relationship is enforced through foreign key constraints, so a CallRecords row can never reference a non-existent Call, Customer, Category, or Reviewer.

## Data Consistency

Controlled vocabularies (Categories, Subcategories, Sentiments, Dispositions) are centralized in lookup tables so identical concepts are represented identically platform wide.

### Auditability

Timestamps, reviewer identity, and edited-field tracking are first-class attributes rather than an afterthought (see Section 15).

### Scalability

Surrogate keys, selective indexing, and a separated analytics table let the schema support millions of calls without redesign (see Section 17).

### Security

Sensitive data is isolated by client company and governed through Role-Based Access Control (see Section 16).

### Performance

Indexes are aligned to primary access patterns: customer lookup, review queue processing, ticket search, and category filtering (see Section 12).

### Future Extensibility

New reference values, review levels, or AI model outputs can be added by extending lookup tables or adding nullable columns, without breaking existing data.

## 3. High-Level Data Flow

The platform processes each completed call through a fixed sequence of stages. The schema is designed so every stage writes to, or reads from, a clearly defined set of tables, keeping the flow traceable end to end.

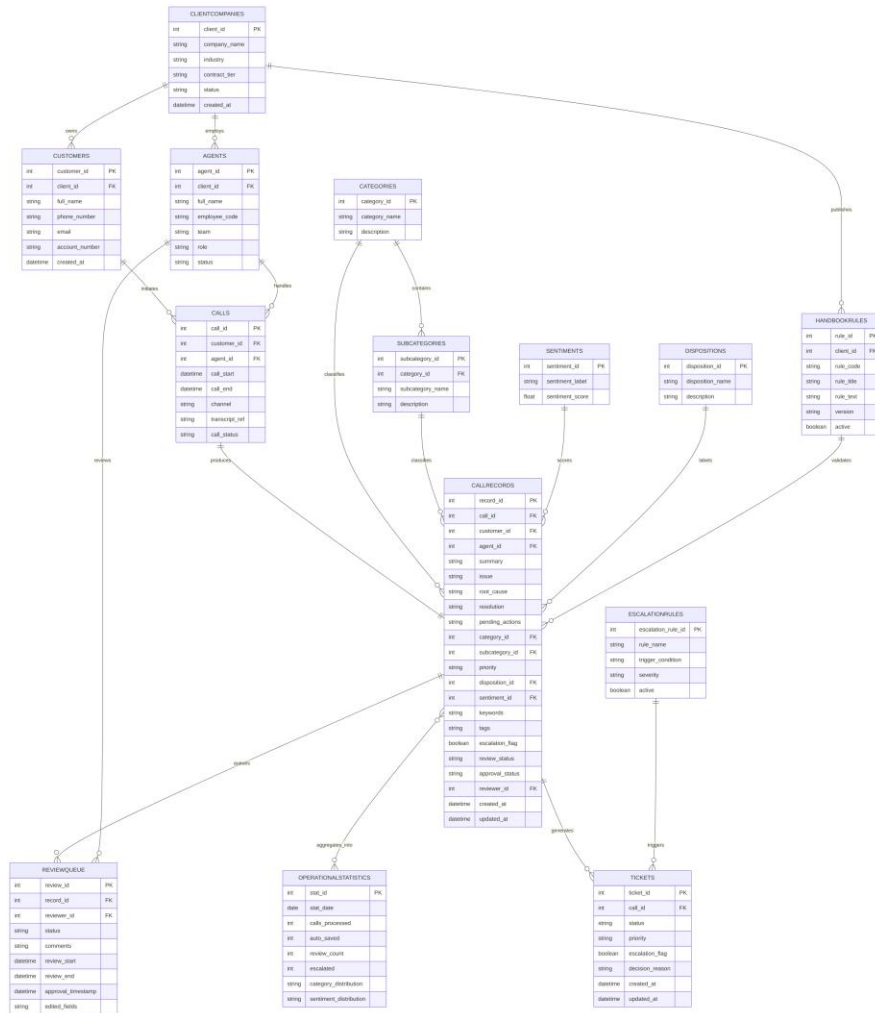
- 1. Transcript Ingestion
- 2. Customer Lookup
- 3. Documentation Retrieval
- 4. AI Extraction
- 5. Classification
- 6. Business Rule Validation
- 7. Human Review
- 8. Final Record Storage
- 9. Operational Analytics

### Stage-to-Table Mapping

| Stage                    | Primary Tables Involved                             |
|--------------------------|-----------------------------------------------------|
| Transcript Ingestion     | Calls                                               |
| Customer Lookup          | Customers, ClientCompanies                          |
| Documentation Retrieval  | HandbookRules                                       |
| AI Extraction            | CallRecords                                         |
| Classification           | Categories, Subcategories, Sentiments, Dispositions |
| Business Rule Validation | EscalationRules, Tickets                            |
| Human Review             | ReviewQueue, Agents                                 |
| Final Record Storage     | CallRecords                                         |
| Operational Analytics    | OperationalStatistics                               |

## 4. Entity Relationship Diagram

The diagram below shows all fourteen entities in the schema and their relationships.



## 5. Entity Descriptions

This section documents every entity in the schema: purpose, columns, keys, constraints, indexes, relationships, business rules, and a representative example record.

### 5.1 ClientCompanies

**Stores the organizations (contact centers or brands) that subscribe to the After-Call Automation Platform.**

Top-level tenant entity. Every Customer, Agent, and HandbookRule is scoped to a client company, enabling isolated multi-tenant operation.

| Column        | Data Type      | Null | Default           | Key |
|---------------|----------------|------|-------------------|-----|
| client_id     | INT (Identity) | No   | Auto-generated    | PK  |
| company_name  | VARCHAR(200)   | No   | None              | -   |
| industry      | VARCHAR(100)   | Yes  | Not Stated        | -   |
| contract_tier | VARCHAR(50)    | No   | Standard          | -   |
| status        | VARCHAR(20)    | No   | Active            | -   |
| created_at    | DATETIME       | No   | Current timestamp | -   |

|                           |                                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | client_id                                                                                                                            |
| <b>Foreign Keys</b>       | None                                                                                                                                 |
| <b>Unique Constraints</b> | company_name (per active tenant)                                                                                                     |
| <b>Indexes</b>            | IX_ClientCompanies_Status on status                                                                                                  |
| <b>Relationships</b>      | 1:N to Customers, Agents, and HandbookRules.                                                                                         |
| <b>Business Rules</b>     | A client company must be Active before new calls can be processed. Deactivation blocks new intake but never deletes historical data. |

*Example: client\_id: 100 | company\_name: Kalam CX Retail Division | industry: Retail Banking | contract\_tier: Enterprise | status: Active*

### 5.2 Customers

**Stores identifying and contact information for the end customers who place support calls.**

Master record used during Customer Lookup, the first automated step of the after-call workflow.

| Column         | Data Type      | Null | Default           | Key |
|----------------|----------------|------|-------------------|-----|
| customer_id    | INT (Identity) | No   | Auto-generated    | PK  |
| client_id      | INT            | No   | None              | FK  |
| full_name      | VARCHAR(150)   | No   | None              | -   |
| phone_number   | VARCHAR(30)    | Yes  | Not Stated        | -   |
| email          | VARCHAR(150)   | Yes  | Not Stated        | -   |
| account_number | VARCHAR(50)    | Yes  | Not Stated        | -   |
| created_at     | DATETIME       | No   | Current timestamp | -   |

|                    |             |
|--------------------|-------------|
| <b>Primary Key</b> | customer_id |
|--------------------|-------------|

|                           |                                                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Foreign Keys</b>       | client_id references ClientCompanies(client_id)                                                                                                             |
| <b>Unique Constraints</b> | (client_id, account_number), where account_number is not Not Stated                                                                                         |
| <b>Indexes</b>            | IX_Customers_Phone, IX_Customers_Account, IX_Customers_Client                                                                                               |
| <b>Relationships</b>      | N:1 to ClientCompanies; 1:N to Calls.                                                                                                                       |
| <b>Business Rules</b>     | Unmatched lookups store Not Stated rather than an assumed identity. Identity fields are never overwritten by AI extraction — only verified profile updates. |

Example: customer\_id: 58211 | client\_id: 100 | full\_name: Sara Ahmed | phone\_number: +20-100-555-0142 | account\_number: ACC-88421

### 5.3 Agents

**Stores the support agents who handle calls and, where applicable, act as reviewers.**

Represents both front-line staff and the reviewer pool used in the Human-in-the-Loop workflow.

| Column        | Data Type      | Null | Default        | Key |
|---------------|----------------|------|----------------|-----|
| agent_id      | INT (Identity) | No   | Auto-generated | PK  |
| client_id     | INT            | No   | None           | FK  |
| full_name     | VARCHAR(150)   | No   | None           | -   |
| employee_code | VARCHAR(30)    | No   | None           | -   |
| team          | VARCHAR(100)   | Yes  | Not Stated     | -   |
| role          | VARCHAR(50)    | No   | Agent          | -   |
| status        | VARCHAR(20)    | No   | Active         | -   |

|                           |                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | agent_id                                                                                                                        |
| <b>Foreign Keys</b>       | client_id references ClientCompanies(client_id)                                                                                 |
| <b>Unique Constraints</b> | (client_id, employee_code)                                                                                                      |
| <b>Indexes</b>            | IX_Agents_EmployeeCode, IX_Agents_Role                                                                                          |
| <b>Relationships</b>      | N:1 to ClientCompanies; 1:N to Calls (as handler); 1:N to ReviewQueue (as reviewer).                                            |
| <b>Business Rules</b>     | Only Reviewer or Supervisor roles may be assigned in ReviewQueue.reviewer_id. Inactive agents cannot take new calls or reviews. |

Example: agent\_id: 340 | client\_id: 100 | full\_name: Omar Nabil | employee\_code: EMP-2245 | role: Reviewer | status: Active

### 5.4 Calls

**Stores metadata for each completed customer support call, including the transcript reference.**

The intake entity that anchors the after-call workflow; parent of exactly one CallRecords row.

| Column      | Data Type      | Null | Default        | Key |
|-------------|----------------|------|----------------|-----|
| call_id     | INT (Identity) | No   | Auto-generated | PK  |
| customer_id | INT            | Yes  | Not Stated     | FK  |

| Column         | Data Type    | Null | Default   | Key |
|----------------|--------------|------|-----------|-----|
| agent_id       | INT          | No   | None      | FK  |
| call_start     | DATETIME     | No   | None      | -   |
| call_end       | DATETIME     | No   | None      | -   |
| channel        | VARCHAR(30)  | No   | Voice     | -   |
| transcript_ref | VARCHAR(300) | No   | None      | -   |
| call_status    | VARCHAR(30)  | No   | Completed | -   |

|                           |                                                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | call_id                                                                                                                 |
| <b>Foreign Keys</b>       | customer_id → Customers(customer_id); agent_id → Agents(agent_id)                                                       |
| <b>Unique Constraints</b> | transcript_ref                                                                                                          |
| <b>Indexes</b>            | IX_Calls_CustomerId, IX_Calls_AgentId, IX_Calls_CallStart                                                               |
| <b>Relationships</b>      | N:1 to Customers (nullable) and Agents; 1:1 to CallRecords.                                                             |
| <b>Business Rules</b>     | Only Completed calls are eligible for AI extraction.<br>transcript_ref must point to an immutable, archived transcript. |

Example: call\_id: 900321 | customer\_id: 58211 | agent\_id: 340 | channel: Voice | call\_status: Completed

## 5.5 CallRecords

**Stores the structured, AI-extracted, and human-reviewed documentation produced for each call — the core business entity of the platform.**

Holds every field produced by AI extraction and classification, plus the review/approval state required by the Human-in-the-Loop workflow. See Section 9 for full field list.

| Column         | Data Type      | Null | Default        | Key |
|----------------|----------------|------|----------------|-----|
| record_id      | INT (Identity) | No   | Auto-generated | PK  |
| call_id        | INT            | No   | None           | FK  |
| customer_id    | INT            | Yes  | Not Stated     | FK  |
| agent_id       | INT            | No   | None           | FK  |
| category_id    | INT            | Yes  | Not Stated     | FK  |
| subcategory_id | INT            | Yes  | Not Stated     | FK  |
| disposition_id | INT            | Yes  | Not Stated     | FK  |
| sentiment_id   | INT            | Yes  | Not Stated     | FK  |
| reviewer_id    | INT            | Yes  | Not Stated     | FK  |

|                           |                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | record_id                                                                                                                                 |
| <b>Foreign Keys</b>       | call_id, customer_id, agent_id, category_id, subcategory_id, disposition_id, sentiment_id, reviewer_id — each references its parent table |
| <b>Unique Constraints</b> | call_id (one CallRecord per Call)                                                                                                         |
| <b>Indexes</b>            | IX_CallRecords_CallId, IX_CallRecords_ReviewStatus, IX_CallRecords_Category, IX_CallRecords_CreatedAt                                     |

|                       |                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Relationships</b>  | 1:1 to Calls; N:1 to Categories, Subcategories, Sentiments, Dispositions; 1:N to Tickets and ReviewQueue.                                                                                                                             |
| <b>Business Rules</b> | No field may be inferred or fabricated — unsupported values store Not Stated. approval_status cannot become Approved until review_status is Reviewed. escalation_flag is set automatically whenever EscalationRules criteria are met. |

*Example: record\_id: 771001 | call\_id: 900321 | issue: Card decline | resolution: Not Stated | review\_status: Pending | approval\_status: Pending*

## 5.6 Tickets

**Stores the follow-up work item created when a call requires action beyond the call itself, including the Decision Engine's reasoning.**

Records the outcome of Business Rule Validation — status, priority, and a decision\_reason explaining the routing decision.

| Column          | Data Type      | Null | Default           | Key |
|-----------------|----------------|------|-------------------|-----|
| ticket_id       | INT (Identity) | No   | Auto-generated    | PK  |
| call_id         | INT            | No   | None              | FK  |
| status          | VARCHAR(30)    | No   | Open              | -   |
| priority        | VARCHAR(20)    | No   | Normal            | -   |
| escalation_flag | BOOLEAN        | No   | False             | -   |
| decision_reason | VARCHAR(300)   | No   | None              | -   |
| created_at      | DATETIME       | No   | Current timestamp | -   |
| updated_at      | DATETIME       | No   | Current timestamp | -   |

|                           |                                                                                                                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | ticket_id                                                                                                                                                                                                                                         |
| <b>Foreign Keys</b>       | call_id references Calls(call_id)                                                                                                                                                                                                                 |
| <b>Unique Constraints</b> | None                                                                                                                                                                                                                                              |
| <b>Indexes</b>            | IX_Tickets_Status, IX_Tickets_CallId, IX_Tickets_Priority                                                                                                                                                                                         |
| <b>Relationships</b>      | N:1 to Calls/CallRecords; N:1 (optional) to EscalationRules.                                                                                                                                                                                      |
| <b>Business Rules</b>     | decision_reason must always state a human-readable explanation (e.g. High Priority, Negative Sentiment, Incomplete Information, Policy Sensitive Case, Customer History). A ticket cannot close while its CallRecords approval_status is Pending. |

*Example: ticket\_id: 55010 | call\_id: 900321 | status: Open | priority: High | decision\_reason: Negative Sentiment*

## 5.7 ReviewQueue

**Stores every human review task, tracking who reviewed a record, what was changed, and when it was approved.**

Implements the mandatory Human-in-the-Loop control point. No CallRecords row may reach final approval without a matching Approved entry here.

| Column      | Data Type      | Null | Default        | Key |
|-------------|----------------|------|----------------|-----|
| review_id   | INT (Identity) | No   | Auto-generated | PK  |
| record_id   | INT            | No   | None           | FK  |
| reviewer_id | INT            | No   | None           | FK  |



| Column             | Data Type     | Null | Default    | Key |
|--------------------|---------------|------|------------|-----|
| status             | VARCHAR(30)   | No   | Pending    | -   |
| comments           | VARCHAR(1000) | Yes  | Not Stated | -   |
| review_start       | DATETIME      | Yes  | Not Stated | -   |
| review_end         | DATETIME      | Yes  | Not Stated | -   |
| approval_timestamp | DATETIME      | Yes  | Not Stated | -   |
| edited_fields      | VARCHAR(500)  | Yes  | Not Stated | -   |

|                           |                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | review_id                                                                                                                                                     |
| <b>Foreign Keys</b>       | record_id → CallRecords(record_id); reviewer_id → Agents(agent_id)                                                                                            |
| <b>Unique Constraints</b> | None                                                                                                                                                          |
| <b>Indexes</b>            | IX_ReviewQueue_Status, IX_ReviewQueue_RecordId, IX_ReviewQueue_Reviewer                                                                                       |
| <b>Relationships</b>      | N:1 to CallRecords; N:1 to Agents (as reviewer).                                                                                                              |
| <b>Business Rules</b>     | edited_fields must list every field changed from the AI draft, preserving the audit trail. approval_timestamp is populated only once status becomes Approved. |

Example: review\_id: 40007 | record\_id: 771001 | reviewer\_id: 340 | status: Approved | edited\_fields: resolution, tags

## 5.8 HandbookRules

**Stores the documentation, policy, and handbook content used for retrieval during the Documentation Retrieval step.**

The knowledge base referenced when validating AI-extracted information against client-specific policy. Rules are versioned for auditability.

| Column     | Data Type      | Null | Default        | Key |
|------------|----------------|------|----------------|-----|
| rule_id    | INT (Identity) | No   | Auto-generated | PK  |
| client_id  | INT            | No   | None           | FK  |
| rule_code  | VARCHAR(50)    | No   | None           | -   |
| rule_title | VARCHAR(200)   | No   | None           | -   |
| rule_text  | VARCHAR(MAX)   | No   | None           | -   |
| version    | VARCHAR(20)    | No   | 1.0            | -   |
| active     | BOOLEAN        | No   | True           | -   |

|                           |                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | rule_id                                                                                                   |
| <b>Foreign Keys</b>       | client_id references ClientCompanies(client_id)                                                           |
| <b>Unique Constraints</b> | (client_id, rule_code, version)                                                                           |
| <b>Indexes</b>            | IX_HandbookRules_Code, IX_HandbookRules_Active                                                            |
| <b>Relationships</b>      | N:1 to ClientCompanies; logically N:N to CallRecords (resolved through validation, not a junction table). |

|                       |                                                                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Business Rules</b> | Only active rules are used to validate new calls; deactivated versions remain for audit. Changes to rule_text create a new version rather than overwriting history. |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Example: rule\_id: 12 | client\_id: 100 | rule\_code: POL-REFUND-01 | rule\_title: Refund Eligibility Policy | version: 2.1 | active: True

## 5.9 Categories

**Stores the top-level classification values used to categorize a call's subject matter.**

The controlled vocabulary used by the AI Classification stage; parent of Subcategories and referenced directly by CallRecords.

| Column        | Data Type      | Null | Default        | Key |
|---------------|----------------|------|----------------|-----|
| category_id   | INT (Identity) | No   | Auto-generated | PK  |
| category_name | VARCHAR(100)   | No   | None           | -   |
| description   | VARCHAR(300)   | Yes  | Not Stated     | -   |

|                           |                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | category_id                                                                          |
| <b>Foreign Keys</b>       | None                                                                                 |
| <b>Unique Constraints</b> | category_name                                                                        |
| <b>Indexes</b>            | IX_Categories_Name                                                                   |
| <b>Relationships</b>      | 1:N to Subcategories; 1:N to CallRecords.                                            |
| <b>Business Rules</b>     | A category cannot be deleted while active Subcategories or CallRecords reference it. |

Example: category\_id: 4 | category\_name: Billing Dispute | description: Charges questioned by the customer

## 5.10 Subcategories

**Stores the second-level classification values that refine a Categories value.**

Enables finer-grained classification and reporting for category filtering.

| Column           | Data Type      | Null | Default        | Key |
|------------------|----------------|------|----------------|-----|
| subcategory_id   | INT (Identity) | No   | Auto-generated | PK  |
| category_id      | INT            | No   | None           | FK  |
| subcategory_name | VARCHAR(100)   | No   | None           | -   |
| description      | VARCHAR(300)   | Yes  | Not Stated     | -   |

|                           |                                                               |
|---------------------------|---------------------------------------------------------------|
| <b>Primary Key</b>        | subcategory_id                                                |
| <b>Foreign Keys</b>       | category_id references Categories(category_id)                |
| <b>Unique Constraints</b> | (category_id, subcategory_name)                               |
| <b>Indexes</b>            | IX_Subcategories_CategoryId                                   |
| <b>Relationships</b>      | N:1 to Categories; 1:N to CallRecords.                        |
| <b>Business Rules</b>     | A subcategory must always belong to a valid, active category. |

Example: subcategory\_id: 21 | category\_id: 4 | subcategory\_name: Duplicate Charge

## 5.11 Sentiments

**Stores the controlled set of sentiment labels applied to a call by the AI extraction pipeline.**

A lookup of sentiment classifications (e.g. Positive, Neutral, Negative) with a numeric score, used for both classification and reporting.

| Column          | Data Type      | Null | Default        | Key |
|-----------------|----------------|------|----------------|-----|
| sentiment_id    | INT (Identity) | No   | Auto-generated | PK  |
| sentiment_label | VARCHAR(30)    | No   | None           | -   |
| sentiment_score | DECIMAL(4,2)   | Yes  | Not Stated     | -   |

|                           |                                                                  |
|---------------------------|------------------------------------------------------------------|
| <b>Primary Key</b>        | sentiment_id                                                     |
| <b>Foreign Keys</b>       | None                                                             |
| <b>Unique Constraints</b> | sentiment_label                                                  |
| <b>Indexes</b>            | IX_Sentiments_Label                                              |
| <b>Relationships</b>      | 1:N to CallRecords.                                              |
| <b>Business Rules</b>     | sentiment_score, when present, must fall between -1.00 and 1.00. |

*Example: sentiment\_id: 3 | sentiment\_label: Negative | sentiment\_score: -0.62*

## 5.12 Dispositions

**Stores the controlled set of outcome labels describing how a call was ultimately resolved or left.**

Standardizes call outcome for consistent reporting and analytics across agents and client companies.

| Column           | Data Type      | Null | Default        | Key |
|------------------|----------------|------|----------------|-----|
| disposition_id   | INT (Identity) | No   | Auto-generated | PK  |
| disposition_name | VARCHAR(50)    | No   | None           | -   |
| description      | VARCHAR(300)   | Yes  | Not Stated     | -   |

|                           |                                                                          |
|---------------------------|--------------------------------------------------------------------------|
| <b>Primary Key</b>        | disposition_id                                                           |
| <b>Foreign Keys</b>       | None                                                                     |
| <b>Unique Constraints</b> | disposition_name                                                         |
| <b>Indexes</b>            | IX_Dispositions_Name                                                     |
| <b>Relationships</b>      | 1:N to CallRecords.                                                      |
| <b>Business Rules</b>     | A disposition cannot be removed while referenced by any CallRecords row. |

*Example: disposition\_id: 2 | disposition\_name: Resolved - First Contact*

## 5.13 EscalationRules

**Stores the configurable business conditions under which a call must be escalated.**

Encodes the criteria evaluated during Business Rule Validation — for example high priority combined with negative sentiment — that trigger escalation tickets.

| Column             | Data Type      | Null | Default        | Key |
|--------------------|----------------|------|----------------|-----|
| escalation_rule_id | INT (Identity) | No   | Auto-generated | PK  |

| Column            | Data Type    | Null | Default | Key |
|-------------------|--------------|------|---------|-----|
| rule_name         | VARCHAR(150) | No   | None    | -   |
| trigger_condition | VARCHAR(300) | No   | None    | -   |
| severity          | VARCHAR(20)  | No   | Medium  | -   |
| active            | BOOLEAN      | No   | True    | -   |

|                           |                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | escalation_rule_id                                                                      |
| <b>Foreign Keys</b>       | None                                                                                    |
| <b>Unique Constraints</b> | rule_name                                                                               |
| <b>Indexes</b>            | IX_EscalationRules_Active                                                               |
| <b>Relationships</b>      | 1:N to Tickets.                                                                         |
| <b>Business Rules</b>     | Only active rules are evaluated for new tickets; inactive rules are retained for audit. |

Example: escalation\_rule\_id: 7 | rule\_name: Negative Sentiment High Priority | severity: High | active: True

### 5.14 OperationalStatistics

**Stores aggregated, pre-computed reporting metrics used for operational analytics dashboards.**

A denormalized reporting table populated by scheduled jobs that summarize CallRecords, Tickets, and ReviewQueue activity into daily/monthly rollups.

| Column                        | Data Type      | Null | Default        | Key |
|-------------------------------|----------------|------|----------------|-----|
| stat_id                       | INT (Identity) | No   | Auto-generated | PK  |
| stat_date                     | DATE           | No   | None           | -   |
| calls_processed               | INT            | No   | 0              | -   |
| auto_saved                    | INT            | No   | 0              | -   |
| review_count                  | INT            | No   | 0              | -   |
| escalated                     | INT            | No   | 0              | -   |
| category_distribution (JSON)  | VARCHAR(MAX)   | Yes  | Not Stated     | -   |
| sentiment_distribution (JSON) | VARCHAR(MAX)   | Yes  | Not Stated     | -   |

|                           |                                                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Key</b>        | stat_id                                                                                                                    |
| <b>Foreign Keys</b>       | None (aggregated from CallRecords, Tickets, ReviewQueue by scheduled jobs)                                                 |
| <b>Unique Constraints</b> | stat_date (one rollup row per day)                                                                                         |
| <b>Indexes</b>            | IX_OperationalStatistics_Date                                                                                              |
| <b>Relationships</b>      | Logically aggregates many CallRecords rows per date; no enforced FK since the relationship is computed, not transactional. |
| <b>Business Rules</b>     | Daily rollups are recomputed if any CallRecords row for that date changes approval_status after initial aggregation.       |

|  |                                                                                                            |
|--|------------------------------------------------------------------------------------------------------------|
|  | monthly_statistics values are derived by summing<br>daily_statistics rows, not recalculated independently. |
|--|------------------------------------------------------------------------------------------------------------|

*Example: stat\_id: 900 | stat\_date: 2026-07-01 | calls\_processed: 1,204 | auto\_saved: 812 | review\_count: 392 | escalated: 46*

## 6. Relationships

### One-to-One

Calls to CallRecords: exactly one structured record is produced per completed call, and a CallRecords row cannot exist without its parent Calls row.

### One-to-Many

- ClientCompanies to Customers, Agents, and HandbookRules.
- Customers to Calls.
- Agents to Calls (as handling agent) and to ReviewQueue (as reviewer).
- CallRecords to Tickets and to ReviewQueue.
- Categories to Subcategories.

### Many-to-One

From the child's perspective, CallRecords to Categories, Subcategories, Sentiments, and Dispositions are each many-to-one: many call records may share the same classification value.

### Cascade, Delete & Update Behavior

- Delete behavior is restrictive by default — a parent row cannot be deleted while child rows reference it, preventing orphans.
- Soft deactivation (a status/active flag) is used instead of physical deletion for ClientCompanies, Agents, HandbookRules, and EscalationRules.
- Update behavior cascades on primary key changes only in rare key-migration cases; surrogate keys are otherwise immutable.
- CallRecords, Tickets, and ReviewQueue rows are never deleted — correction happens through the review workflow, which appends audit history.

### Relationship Cardinality

| Parent          | Child         | Cardinality   |
|-----------------|---------------|---------------|
| ClientCompanies | Customers     | 1:N           |
| ClientCompanies | Agents        | 1:N           |
| ClientCompanies | HandbookRules | 1:N           |
| Customers       | Calls         | 1:N           |
| Agents          | Calls         | 1:N           |
| Calls           | CallRecords   | 1:1           |
| Categories      | Subcategories | 1:N           |
| Categories      | CallRecords   | 1:N           |
| Subcategories   | CallRecords   | 1:N           |
| Sentiments      | CallRecords   | 1:N           |
| Dispositions    | CallRecords   | 1:N           |
| HandbookRules   | CallRecords   | N:N (logical) |
| CallRecords     | Tickets       | 1:N           |
| CallRecords     | ReviewQueue   | 1:N           |

| Parent          | Child       | Cardinality |
|-----------------|-------------|-------------|
| Agents          | ReviewQueue | 1:N         |
| EscalationRules | Tickets     | 1:N         |

## 7. Constraints & Indexing

### Primary & Foreign Keys

Every table uses a single-column surrogate integer primary key (identity/auto-increment), providing a stable, non-business-meaningful identifier. Foreign keys enforce every parent-child relationship in Section 6, ensuring CallRecords, Tickets, and ReviewQueue rows can never reference non-existent parents.

### Unique Constraints & NOT NULL

Unique constraints prevent duplicate business identities, such as one CallRecords row per Calls row, one HandbookRules version per rule code, and one OperationalStatistics row per calendar date. Fields required for the platform to function (call\_id, agent\_id, priority, review\_status, approval\_status) are NOT NULL. Fields that may be genuinely unknown (customer\_id, root\_cause) are nullable and default to the literal value Not Stated rather than remaining empty.

### CHECK Constraints

- priority must be one of Low, Normal, High, Critical.
- review\_status must be one of Pending, In Review, Reviewed.
- approval\_status must be one of Pending, Approved, Rejected.
- sentiment\_score, when present, must be between -1.00 and 1.00.

### Index Recommendations

| Use Case           | Recommended Index                                       |
|--------------------|---------------------------------------------------------|
| Customer Lookup    | IX_Customers_Phone, IX_Customers_Account                |
| Review Queue       | IX_ReviewQueue_Status, IX_ReviewQueue_Reviewer          |
| Analytics          | IX_OperationalStatistics_Date, IX_CallRecords_CreatedAt |
| Ticket Search      | IX_Tickets_Status, IX_Tickets_Priority                  |
| Transcript Search  | IX_Calls_TranscriptRef                                  |
| Category Filtering | IX_CallRecords_Category (category_id, subcategory_id)   |



## 8. Normalization & Data Integrity

### Normalization (3NF)

Every non-key attribute in each table depends on the whole primary key and nothing but the primary key. Reference values such as category names, sentiment labels, and dispositions are extracted into their own tables rather than repeated as free text on CallRecords.

### Intentional Denormalization

OperationalStatistics is intentionally denormalized. It stores pre-aggregated counts and JSON distribution summaries that could, in principle, be derived on demand — this protects transactional performance by moving expensive aggregation off the live call-processing path onto a scheduled batch process.

### Data Integrity Rules

#### No Orphan Records

Foreign keys and restrictive delete behavior ensure Tickets, ReviewQueue, and CallRecords rows can never exist without a valid parent Calls row.

#### Required Relationships

Every CallRecords row must link to a Calls row and an Agents row; every ReviewQueue row must link to a CallRecords row and a reviewing Agents row.

#### Mandatory Review

A CallRecords row cannot reach Approved unless review\_status is Reviewed and a corresponding Approved ReviewQueue entry exists.

#### Mandatory Audit Trail

created\_at, updated\_at, reviewer\_id, and edited\_fields are populated for every reviewed record.

#### No Fabricated Data

AI extraction is prohibited from inferring facts not present in the transcript or a HandbookRules entry.

#### Unsupported Values Become Not Stated

Any field that cannot be confidently determined is stored as the literal text Not Stated rather than an empty string, a guess, or an ambiguous NULL.

## 9. Audit Trail & Security

### Audit Trail

#### Original AI Draft

Initial values from the AI extraction pipeline are preserved in CallRecords at created\_at, before any reviewer edits.

#### Human Edits

Any field changed during review is listed in ReviewQueue.edited\_fields, showing the precise difference between the AI draft and the human-approved version.

#### Reviewer & Approval History

ReviewQueue.reviewer\_id records who reviewed the record; status and approval\_timestamp capture the outcome and timing of every decision, including rejections.

#### Timestamps

created\_at / updated\_at on CallRecords, plus review\_start / review\_end on ReviewQueue, provide a full timeline for each record.

#### Decision History

Tickets.decision\_reason preserves the Decision Engine's rationale at the moment of routing, independent of any later human override.

### Security Considerations

#### Role-Based Access Control

Access to Customers, CallRecords, and ReviewQueue is governed by role, so agents, reviewers, supervisors, and analytics users see only appropriate data.

#### Data Privacy

Customer identifying fields (full\_name, phone\_number, email, account\_number) are isolated in Customers so sensitive personal data can be controlled and audited separately from operational data.

#### Least Privilege

Write access to reference tables (Categories, Dispositions, EscalationRules) should be restricted to platform administrators.

#### Encryption

Sensitive columns, particularly customer contact details, should be encrypted at rest, and connections should use encryption in transit.

#### Audit Logs

Database-level audit logging should complement the application-level audit trail, capturing direct data access outside the normal workflow.

## 10. Scalability & Future Enhancements

### Scalability Considerations

#### Millions of Calls

Calls and CallRecords use narrow, indexed integer keys and are designed for partitioning by date or client company as volume grows.

#### Thousands of Customers

Customers is indexed on phone\_number and account\_number to keep Customer Lookup fast regardless of total customer count.

#### Growing Handbook

HandbookRules supports versioning, letting the knowledge base expand without invalidating historical validation decisions.

#### Large Analytics Datasets

OperationalStatistics absorbs analytics load through pre-aggregation, keeping dashboard queries fast even at hundreds of millions of transactional rows.

#### Future AI Models

Nullable, independently indexed classification columns allow new or improved AI models to populate the same structure without schema changes.

### Future Enhancements

- Version history for CallRecords, preserving every reviewer edit as a discrete versioned snapshot.
- Multi-language support for transcripts, summaries, and handbook content.
- Multiple review levels, allowing supervisor sign-off for high-severity cases.
- Quality scoring of AI extraction accuracy, tracked per record and per model version.
- A feedback loop that routes reviewer corrections back into model evaluation datasets.
- Model performance tracking, correlating outcomes with the AI model version that produced the extraction.

## 11. Appendix

### 11.1 Entity Summary

| Entity                | Purpose                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| ClientCompanies       | Stores the organizations (contact centers or brands) that subscribe to the After-Call Automation Platform.                               |
| Customers             | Stores identifying and contact information for the end customers who place support calls.                                                |
| Agents                | Stores the support agents who handle calls and, where applicable, act as reviewers.                                                      |
| Calls                 | Stores metadata for each completed customer support call, including the transcript reference.                                            |
| CallRecords           | Stores the structured, AI-extracted, and human-reviewed documentation produced for each call — the core business entity of the platform. |
| Tickets               | Stores the follow-up work item created when a call requires action beyond the call itself, including the Decision Engine's reasoning.    |
| ReviewQueue           | Stores every human review task, tracking who reviewed a record, what was changed, and when it was approved.                              |
| HandbookRules         | Stores the documentation, policy, and handbook content used for retrieval during the Documentation Retrieval step.                       |
| Categories            | Stores the top-level classification values used to categorize a call's subject matter.                                                   |
| Subcategories         | Stores the second-level classification values that refine a Categories value.                                                            |
| Sentiments            | Stores the controlled set of sentiment labels applied to a call by the AI extraction pipeline.                                           |
| Dispositions          | Stores the controlled set of outcome labels describing how a call was ultimately resolved or left.                                       |
| EscalationRules       | Stores the configurable business conditions under which a call must be escalated.                                                        |
| OperationalStatistics | Stores aggregated, pre-computed reporting metrics used for operational analytics dashboards.                                             |

### 11.2 Glossary

| Term              | Definition                                                                                                     |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| Auto Save         | A business decision indicating a record is eligible for final storage without further human review.            |
| Route to Review   | A business decision sending a record into the ReviewQueue for human validation.                                |
| Escalation        | A business decision indicating a record requires elevated handling due to severity or policy sensitivity.      |
| Not Stated        | The controlled value stored when information cannot be confidently determined from the transcript or handbook. |
| Human-in-the-Loop | A workflow design requiring human review and approval before a system-generated output becomes final.          |

| Term            | Definition                                                                                               |
|-----------------|----------------------------------------------------------------------------------------------------------|
| Decision Engine | The business rule component that evaluates a record and produces a routing decision and decision_reason. |

### 11.3 Abbreviations

| Abbreviation | Meaning                    |
|--------------|----------------------------|
| ERD          | Entity Relationship Design |
| PK           | Primary Key                |
| FK           | Foreign Key                |
| 3NF          | Third Normal Form          |
| RBAC         | Role-Based Access Control  |
| AI           | Artificial Intelligence    |